

CURRICULUM VITAE

Vladimir Vladimirovich Baryshev

[Email](#)

[LinkedIn](#)

[ORCID](#)



EDUCATION

Bachelor: General Biology

Specialization: Molecular Biology and Genetics

North-Caucasus Federal University — Stavropol, Russia

GPA: 4.35

Master: Neurotechnology with the basics of biomaterials science

Sirius — Sochi

First course: In progress

PROFESSIONAL DEVELOPMENT (ADVANCED TRAINING)

Technologies for Controlling Biological Properties: Practical Course in Bioinformatics and Molecular Biology

Institution: Novosibirsk State University

Completed: 2025 (9 months)

Molecular Genetic Methods for Investigating Living Systems

Institution: North-Caucasus Federal University

Completed: 2023 (6 months)

P.S. At the time of application, I have not yet received the official certificates confirming the completion of these programs. However, I will be able to provide them by the start of the academic year.

SCHOOLS & WORKSHOPS

- Sirius Neurotechnology School (focus: biomaterials and neurointerfaces) — March 2025 (*Russian*): [link](#)
- X International School for Young Scientists (*Active and Passive Methods of Brain Research, HSE*) — Nov 2024 (*English*): [link](#)
- BCI & Neurotechnology Spring School (*g.tec*) — 2025 (*English*): [link](#)

- Neuromatch Academy (Computational Neuroscience Track) — Summer 2025 (English): [link](#)
- Summer School “Neurobiology of Attention and Memory” named after O.S. Vinogradova — Aug 2025 (Russian): [link](#)
- *New Generation Neurointerfaces: Prospects for Practical Application (HSE)* — May 2025 (Russian): [link](#)

PUBLICATION

Lyakhov P. A., Lyakhova U. A., Baboshina V. A., Baryshev V. V., Nagornov N. N.

Detection of attention state in children with autism spectrum disorder based on neural network classification of electroencephalograms.

Vestnik of Saint Petersburg University. Applied Mathematics. Computer Science. Control Processes, 2025, vol. 21, iss. 1, pp. 92–111. [<https://doi.org/10.21638/spbu10.2025.107>]

Link: [ResearchGate](#)

Q3 Scopus

RESEARCH AND PROJECT EXPERIENCE

My research journey began with developing a software tool for morphological processing of histological images, where I was responsible for implementing a complete image analysis pipeline in Python. This work contributed to my thesis and helped me strengthen my skills in algorithm design, morphological filtering, and biomedical image enhancement.

Another milestone was a long-term independent EEG project focused on classifying attention states in children with Autism Spectrum Disorder (ASD) using neural networks. I initiated and led the entire research cycle — from problem definition and literature review to data preprocessing, model training, and paper writing. The project involved collaboration with medical institutions and provided me with critical experience in applied neuroscience and interdisciplinary coordination.

Currently, I lead a small team working on motor imagery EEG decoding, where I design neural network architectures and coordinate research tasks. We are also preparing a narrative review article covering motor-related BCI paradigms. In parallel, I am exploring metric learning approaches, particularly embedding EEG signals via triplet loss variations to improve model generalization across subjects.

GRANTS

Software developed as part of my diploma thesis was supported by a grant from the Russian Science Foundation (project 23-71-10013).

Title: "Software for Morphological Processing of Histological Slice Images and Object Segmentation"

The software, developed in Python using OpenCV, NumPy, Matplotlib, and PyQt, enables complex processing of histological images through color correction (Gray World), channel normalization, threshold binarization, and morphological operations (erosion, dilation, opening, closing). It prepares data for downstream morphometric and diagnostic analysis.

ACADEMIC INTERESTS

I am passionate about neuroscience, neurotechnology, and the interplay between biology, AI, and materials science. I am particularly interested in:

- Brain-Computer Interfaces (BCI) based on functional nanoparticles
- Non-invasive stimulation via piezoelectric mechanisms
- Reverse piezoelectric detection for feedback
- Magnetic guidance of nanoparticles
- EEG decoding for motor and attention states
- Metric learning in neural networks for EEG embedding
- Integration of neurotech, machine learning, and biophysics
- Bioinformatics and neural signal processing
- Interdisciplinary methodologies in neurobiology

SOFTWARE & TECHNICAL SKILLS

- Python (PyTorch, NumPy, MNE, OpenCV, Matplotlib)
- GUI development (PyQt)
- Scientific image processing, EEG preprocessing, neural networks
- Platforms: Git, Linux (basic)

SCIENCE COMMUNICATION & JOURNALISM

Alongside my academic and research work, I am actively engaged in science journalism and communication. As a staff science writer for [Science Mail](#) I have authored over 800 popular science articles, covering a wide range of topics including neuroscience, genetics, physics, artificial intelligence, and biomedicine. This work has strengthened my ability to translate complex research into accessible language, and deepened my interest in interdisciplinary approaches that connect scientific knowledge with real-world impact.

LANGUAGE SKILLS

- Russian: Native
- English: Intermediate–Upper Intermediate (B2)